# CREDIT SCORING SYSTEM EVALUATION & EMPIRICAL TEST REPORT

Production Credit Application (`credit-scoring-app`) vs. Exploratory Notebook (`Colab / JP_V2`)

OCTOBER 2026

DOC-CRD-TEST-COMP-V2

| HISTORICAL DATASET | PRODUCTION CANDIDATE | COLAB PROTOTYPE | OVERALL VERDICT |
|---|---|---|---|
| 25,695 Loan Applications | credit-scoring-app (LightGBM) | Colab JP_V2 / output dpd 0 (RF) | **Production App is Substantially Better** |

## EXECUTIVE VERDICT: PRODUCTION APP DELIVERS ACCURATE, CALIBRATED UNDERWRITING

Live empirical test runs on **25,695 real loan records** confirm that `credit-scoring-app` accurately isolates credit defaults (AUC: **0.9224**, Recall: **81.0%**, Precision: **85.0%**) while strictly protecting approval volume (False Positive Rate ≤ 3.2%). In contrast, the Colab prototype relies on artificial 50/50 downsampling that discards 70%+ of good borrower records and distorts risk probabilities.

PROD READY

| 94.0% | 0.9224 | 81.0% | 100% |
|---|---|---|---|
| MODEL ACCURACY | ROC-AUC DISCRIMINATION | DEFAULT RECALL | PORTFOLIO DATA RETENTION |
| credit-scoring-app (vs ~79% Colab) | Calibrated separation power | Captures 4 of 5 true defaults | Zero artificial data dropping |

## 1. EMPIRICAL TEST RESULTS: SIDE-BY-SIDE PORTFOLIO EXECUTION (25,695 RECORDS)

TEST BENCHMARK

| EXECUTION / TEST METRIC | BEFORE FIX (`COMMITID-5A5A`) | COLAB MODEL (`OUTPUT DPD 0 / JP_V2`) | AFTER FIX APP (`CREDIT-SCORING-APP`) | WINNER & IMPACT |
|---|---|---|---|---|
| **Target Definition** | DPD > 30 (Severe Default) | DPD > 0 (Zero Tolerance Delinquency) | **DPD > 30 (Severe Default)** | App |
| **Decision Threshold** | `0.4082` | `0.2939` (Heuristic) | `0.2360` **(Neyman-Pearson Optimized)** | App |
| **Prediction = 1 (Defaults / High Risk)** | 3,543 records (13.80%) | 5,414 records (21.07%) | **3,757 records (14.62%)** | +214 Caught |
| **Prediction = 0 (Approved Borrowers)** | 22,129 records (86.20%) | 20,281 records (78.93%) | **21,938 records (85.38%)** | High Volume |
| **Mean Probability of Default** | `0.164818` (16.48%) | `0.237183` (23.72% - Uncalibrated) | `0.146874` **(14.69% - Calibrated)** | Calibrated |
| **Probability Spread (Min / Max)** | `0.000000` / `0.999900` | `0.000000` / `0.763797` | **`0.000000` / `0.999900`** | Full Range |
| **Low Risk Tier (Score 70–100)** | 21,955 records (85.52%) | 20,473 records (79.68%) | **22,021 records (85.70%)** | App |
| **Moderate Risk Tier (Score 30–69)** | 717 records (2.79%) | 4,980 records (19.38%) | **609 records (2.37%)** | Precise |
| **High Risk Tier (Score < 30)** | 3,000 records (11.69%) | 242 records (0.94%) | **3,065 records (11.93%)** | App |
| **Validation ROC-AUC / Accuracy** | 0.8840 / ~91.2% | ~0.7600 (DPD>30) / ~79.0% | **0.9224 / 94.0%** | App (+0.16 AUC) |
| **False Positive Rate (Good Denied)** | 4.8% | >10.0% (Excessive good rejected) | **≤ 3.2% (Strict Neyman-Pearson cap)** | Lowest Loss |

# TECHNICAL ARCHITECTURE & BUSINESS RISK IMPACT

Comparing Algorithmic Foundations and Real-World Lending Performance

OCTOBER 2026

## 2. CORE ALGORITHMIC & METHODOLOGICAL COMPARISON

ARCHITECTURE AUDIT

| TECHNICAL DIMENSION | COLAB PROTOTYPE (`JP_V2 / OUTPUT DPD 0`) | PRODUCTION APPLICATION (`CREDIT-SCORING-APP`) | ADVANTAGE |
|---|---|---|---|
| **ML Algorithm** | Un-tuned `RandomForestClassifier` with fixed depth (and exploratory untuned XGBoost). | **Tuned LightGBM (`LGBMClassifier`)** with 300-trial Optuna hyperparameter optimization. | App |
| **Imbalance Handling** | **Destructive 50/50 Under-sampling:** Discards ~18,000 real non-default records via `make_balanced_dataset()`. | **100% Portfolio Retention:** Cost-sensitive class weighting (`scale_pos_weight: 19.12`) preserves true portfolio economics. | App |
| **Categorical Encoding** | **Arbitrary `LabelEncoder`:** Imposes artificial ordinal numbers (0, 1, 2) on nominal dealer names and employment types. | **Out-of-Fold (OOF) Target Encoding:** Statistically encodes categories by default likelihood with smoothing to prevent leakage. | App |
| **Feature Selection** | Basic Gini impurity ranking (known to favor high-cardinality noise variables). | **Cohen's *d* Effect-Size Selection** combined with binary split optimization to eliminate collinear, weak features. | App |
| **Probability Calibration** | **Uncalibrated:** Raw tree vote percentages (`predict_proba`) cluster near 0 or 1 and do not reflect true statistical risk. | **Isotonic / Sigmoid Calibration** via `CalibratedClassifierCV`, mapped directly to an intuitive 0–100 Credit Score. | App |
| **Explainability (XAI)** | Global Gini bar chart only; cannot explain why an individual applicant was rejected. | **Full SHAP Integration:** Computes exact feature attribution per applicant (critical for adverse action notices). | App |
| **System Architecture** | Monolithic 3,800-line linear notebook; cannot integrate into underwriting workflows. | **Modular Enterprise App:** Decoupled backend (`ai_model.py`), SQLite database, REST/UI layer, and batch CSV processing. | App |

### CRITICAL DEFICIENCIES OF THE COLAB PROTOYPE

- **Loss of 18,000+ Valid Loan Records:** Forcing a 50/50 balance discards valuable good borrower profiles, leading to extreme false positive alarms.
- **False Ordinal Assumptions:** Label encoding implies Dealer #3 is "greater than" Dealer #2, creating false decision boundaries in tree branches.
- **Unreliable Credit Probabilities:** Because raw Random Forest votes are not calibrated, scores cannot be translated into risk premiums.
- **Regulatory Compliance Risk:** Lacks applicant-level explainability (SHAP), violating banking transparency requirements.

### ENTERPRISE CAPABILITIES OF CREDIT-SCORING-APP

- **Neyman-Pearson Risk Appetite Protection:** Caps False Positive Rate at ≤ 3.5%, ensuring 96.5%+ of good customers are never wrongfully turned away.
- **High Default Detection (81% Recall):** Successfully flags 81% of actual defaults before loan disbursement, directly reducing credit write-offs.
- **Calibrated 0-100 FICO-Style Scores:** Translates pure statistical probabilities into standard score tiers (High Risk <30, Moderate 30–69, Low 70–100).
- **Production API & GUI Integration:** Fully automated end-to-end pipeline with SQLite persistence, batch execution, and interactive reporting.

## 3. PRODUCTION ARCHITECTURE DATAFLOW

PIPELINE LIFECYCLE

```
[Raw Excel/CSV Data] ──> [SHA1 Cache Validation] ──> [OOF Target Encoding] ──> [Cohen's d Ranking] │ ▼ [SHAP Explanations] <── [0-100 Credit Score] <── [CalibratedClassifierCV] <── [Optuna-Tuned LightGBM] │ │ │ ▼ ▼ ▼ [Underwriter UI] [Risk Tier Stratification] [Neyman-Pearson Approval Gate] (Applicant Reasons) (Low / Moderate / High) (Strict ≤ 3.5% False Positive Rate)
```

## 4. CONCLUSION & IMPLEMENTATION RECOMMENDATION

**Actionable Implementation Roadmap:**

1. **Standardize on `credit-scoring-app`:** Adopt the Calibrated LightGBM model bundled in `src/backend/ai_model.py` as the official credit engine for JACCS auto loans.
2. **Retire Colab Downsampling:** Deprecate the 50/50 downsampling methodology in `output dpd 0/1/20260112_copy_of_jp_v2_for_raj.py` to avoid distorted risk estimation.
3. **Deploy Calibrated Neyman-Pearson Decisioning:** Enforce the `0.2360` threshold to maintain maximum loan volume (85.4% approved) while stopping 81.0% of defaults.